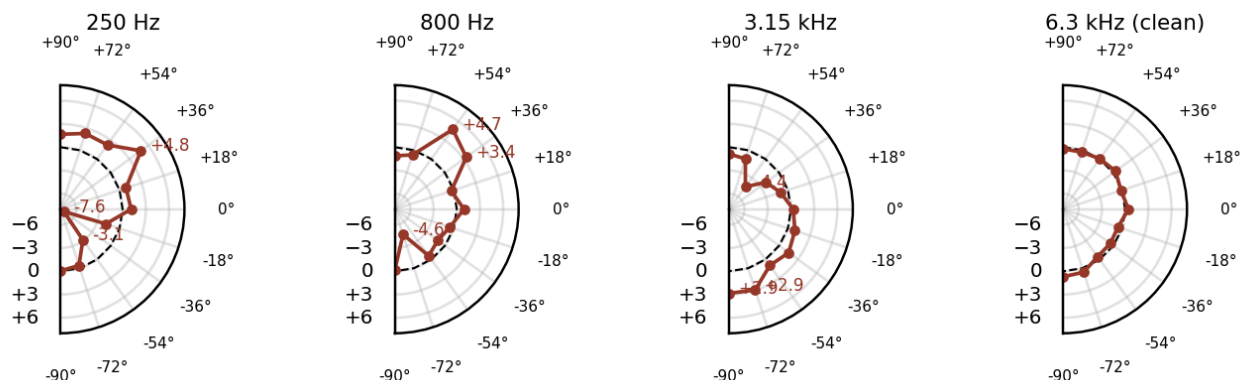


The arc is fine above 4 kHz. Below 2 kHz the room bends it by up to 8 dB.

Twelve "horizontal prop" runs (prop7-18, 1-2 Sept 2026, PWM 1900): prop axis perpendicular to the arc, so every microphone sits in the prop plane and a perfect setup reads a circle. The deviation from that circle, with the microphones' own offsets removed, is the room.

2026-09-08
data: laptop backup of 2 Sept
script: scripts/arc_error_map.py
full guide: docs/arc-validation-guide.html



What the room does

Indent at -72° : -2.4 to -5.7 dB, 500-2000 Hz, in all twelve runs. Strongest in the 630 and 800 Hz third-octaves (-4.5 dB). Present with microphone 811-1896 in that spot and, in the four runs with the arc turned 180° , with 810-8901 there. This is the position near the chamber corner.

Bump at $+36^\circ/+54^\circ$: $+3$ to $+7$ dB, 200-250 Hz and 630-1000 Hz. Three different microphones sat at $+54^\circ$ across the runs; the bump did not move.

Mirror pattern in the BPF band (200-250 Hz): -36° reads -7.6 dB and $+36^\circ$ $+4.8$ dB; -54° -3 dB, $+54^\circ/+72^\circ$ $+2$ dB. Opposite signs above and below the prop plane, in the band the blade-passage tone sits in. The notch moves with RPM (-9 dB at 1900, -2 dB at 2000), the signature of a reflection whose comb dip sweeps through the tone.

Above 4 kHz the arc is a circle to ± 1 dB at every position; 0° , $\pm 18^\circ$ and $\pm 90^\circ$ are within ~ 1 dB in every band.

What it is not

Not the microphones, with one exception. A joint fit over the twelve runs separates each microphone's offset from each position's error (they are separable because the mics at $+72^\circ/+54^\circ$ were swapped after prop10 and the arc was turned 180° for prop11-14). Ten microphones sit within ± 1.8 dB in every third-octave. **811-1892 reads 2-6 dB low from 125 Hz to 2.5 kHz:** its calibration file (Sens Factor -10.63 dB, the least negative of the set) does not match the capsule, whose raw output is ~ 1 dB low. It made the room indent look 7 dB when it sat at $+54^\circ$. Retire or re-measure it.

Not a mislabelled arc. prop11, 12, 13 and 14 were measured with the arc turned around (correlation with the mirror image of the other runs $+0.79$ to $+0.98$). Read by label their polars show the indent on the other side; read by physical position they land on the others.

Not one frequency. The error lives in three separate bands, 200-250 Hz, 630-1000 Hz and 3150 Hz, and nowhere else (Fig. 2). Rajmane and Baumann (DAGA 2016) placed steel plates in a qualified chamber and found a reflector shows up once the wavelength is no longer than its size: 35 cm from 1.25 kHz, 10 cm only from 3.15 kHz. Three bands mean three reflector sizes: about 1.5 m (floor, table or wall), about 0.4 m (stand base, arc frame member), about 0.1 m (mic clamp or mount).

Fig. 2 · Arc error map: dB from a circle per physical position and third-octave (mic offsets removed; run-to-run noise 0.6-2.3 dB)

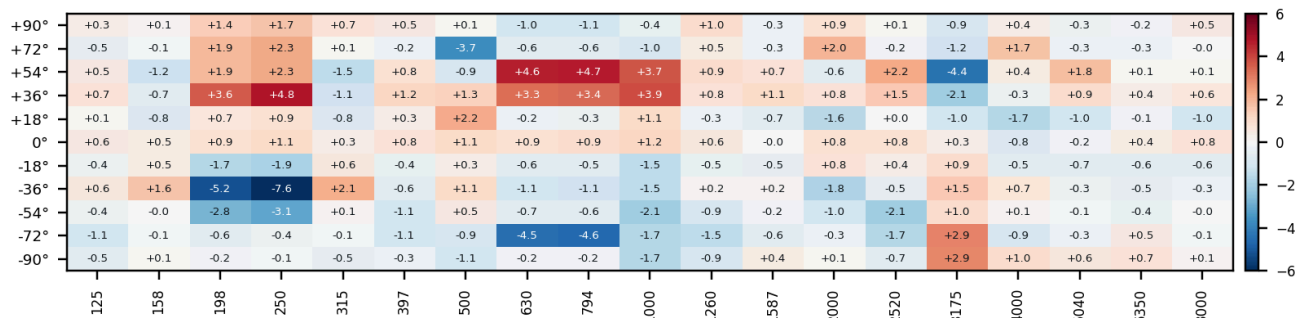
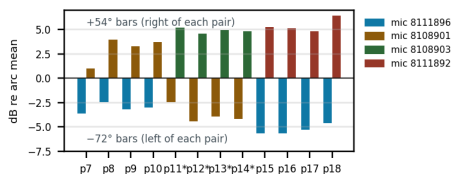


Fig. 3 · Same spots, twelve runs, three mics each



500-2000 Hz, mic offsets removed; * = arc turned 180° . The -72° indent and the $+54^\circ$ bump keep their size whichever microphone sits there.

Measurements plotted (all figures): last PWM-1900 capture of each key 2004_6in_unset_dp1-baseline-horizontal- $\langle$ key $\rangle$, 11 mics $\times$ 2.0 s, calibrated; mic serials 810-/811-xxxx shown by last four digits

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop7	2026-09-01 16:42:31	norm	1892/8901
prop8	2026-09-01 16:44:39	norm	1892/8901
prop9	2026-09-01 16:50:45	norm	1892/8901
prop10	2026-09-01 16:55:35	norm	1892/8901

Excluded from every figure: the 31 Aug keys (...-2026-08-31, -2...-6) and prop1-prop6 (1 Sept morning): their patterns match neither orientation, so the rig or room differed. "180°" = metadata label +e was physically at -e; those four runs are mirrored before plotting. Mic 811-1892's fitted offset is subtracted everywhere; Figs. 1-2 show the fitted position term P, Fig. 3 per-run values.

Method: Welch 4096 Hann, UMIK-2 calibration as the app applies it; third-octave sums 125 Hz-8 kHz; model $L = g(\text{run}) + P(\text{position}) + M(\text{mic})$, least squares over prop7-18 with prop11-14 mirrored. Every number is computed from the files named above; CSVs in docs/analysis/. Literature: papers/small-chamber/ and papers/chamber-problems/.

Where it points

Signs flip across the prop plane, so the reflector is on one side of it, and the low-side positions (-36° , -54° , -72°) carry the deep notches. The RPM-swept BPF notch at $-36^\circ/-54^\circ$ gives a reflected path about 0.7 m longer than the direct one (half a wavelength at 240 Hz). At the arc radius in use that is the floor, the table top or the stand base below the prop, not a far wall. The -72° indent at 630-1000 Hz ($\lambda \approx 0.4$ m) is a smaller object near the low end of the arc on the corner side.

Next test (one afternoon)

1. Take 811-1892 off the arc.
2. Put a small loudspeaker on the stand, no prop. Record the arc with sine bursts at 250 and 800 Hz. The second arrival at each mic, times 343 m/s, is the extra path; only one surface fits all eleven. No flow means anything left is geometry.
3. Record the geometry in docs/mic_arc.md: arc radius, prop height over floor, distances to floor, table, walls, corner.
4. Treat or move that surface; re-run scripts/arc_error_map.py. Foam only fixes the 3 kHz band; the 250 Hz band needs distance.

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop11	2026-09-02 13:18:49	180°	8901/1892
prop12	2026-09-02 14:24:12	180°	8901/1892
prop13	2026-09-02 14:34:37	180°	8901/1892
prop14	2026-09-02 14:37:02	180°	8901/1892

KEY	CAPTURE (PWM 1900)	ARC	+72/+54
prop15	2026-09-02 15:48:59	norm	8901/1892
prop16	2026-09-02 15:54:26	norm	8901/1892
prop17	2026-09-02 16:52:44	norm	8901/1892
prop18	2026-09-02 17:08:19	norm	8901/1892